CYBER DECEPTION FOR CRITICAL INFRASTRUCTURE RESILIENCY

-To discover a vulnerability and applying patch is long often 5 months or more. So these delays cause significant risks to the organizations while many cyber networks are operational.

-First thing they focused on is **“how can defender use the network of decoys along with the real network to introduce mistrust”**.

-And another thing is whether to spend or save resources resources(number of decoys).

-**Dynamic Deception System (DDS)** with **Software Defined Networking (SDN)**.

- **SDN** works like a smart traffic controller — it can tell apart normal traffic from suspicious/malicious traffic.
- **Benign traffic** → goes straight to our real critical infrastructure systems.
- **Malicious traffic** → doesn’t get blocked immediately. Instead, SDN **reroutes it into our deception server**.
- The **deception server** is a fake environment that looks real, so attackers continue their activities there without realizing.
- This lets us **observe, profile, and study attacker behavior** safely, while protecting the real systems.

-They adressed the challenge of efficiently placing network decoys by predicting the most likely attack path in Multi-Stage Attacks (MSAs). MSAs are cyber security threats where the attack campaign is performed through several attack stages and adversarial lateral movement is one of the critical stages.

-Hackers often move **laterally** inside a network after the first break-in. To stop this they had built a **two-phase defense approach** that mixes **graph analysis** (reactive) and **cyber deception** (proactive).

- **Phase 1 – Predicting the attack path**
  Using **IDS alerts** and **network traffic traces**, we apply a **Hidden Markov Model (HMM)**. This helps us calculate the **most likely route an attacker will take** when moving through the network.
- **Phase 2 – Deploying deception**
  Once we know the likely attack path, we place **decoy nodes (fake systems)** along it. These decoys look real, so when the attacker follows the path, they get trapped in deception instead of reaching critical assets.

Problem Statements & Contributions

1. How to secure a vulnerable network by balancing security cost vs availability cost.
2. How to use a deception framework that considers attacker capabilities to optimize defense resources.
3. How to predict likely attack paths so decoys can be placed to block lateral movement.
4. How to build a deception-based system that profiles attacker strategies, reduces their success, and increases their resource usage.

This paper is about protecting **critical infrastructure** from cyberattacks using **cyber deception** and they had built a **Dynamic Deception System (DDS)** that uses **SDN** to separate normal traffic from malicious traffic and redirect attackers into **fake decoy systems** they used **graph analysis and Hidden Markov Models (HMM)** to **predict the attacker's next move** in multi-stage attacks. Based on this prediction, they were placing **decoys along likely attack paths** to block lateral movement. The system also studies attacker behavior to **adapt defenses dynamically**. To put it more in precise it **tricks attackers, wastes their time, and protects real systems**.